Kumari Akanksha

Computational Imaging · Scientific Machine Learning · Deep Learning for Biological Systems

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Research Interests

Computational microscopy · biological image analysis · multi-object tracking & re-identification · spatiotemporal modeling · instance segmentation · weakly supervised learning · interpretable AI · representation learning · image-based phenotyping · quantitative biology · reproducible research pipelines

Education

National Cheng Kung University (NCKU) — Tainan, Taiwan
M.S. in Biomedical Engineering — GPA: 4.24/4.30 ***(top of cohort)****, expected Feb 2027* — *Feb 2025 – Present*

Delhi Skill and Entrepreneurship University (DSEU) — Delhi, India
B.S. in Data Analytics — CGPA: 8.89/10.0 — *Aug 2021 – May 2024*

Research Experience

Graduate Research Assistant, Tu Lab — Feb 2025 – Present
National Cheng Kung University · Advisor: Prof. Ting-Yuan Tu — *Tainan, Taiwan*

- **Live-cell tracking & identity re-identification.** Developing a contrastive-embedding Siamese network for long-term cell identity preservation in dense time-lapse confocal microscopy sequences of 3D cancer spheroids embedded in Type-I collagen matrices; dataset spans 12 spheroid clusters, ∼1,118 frames with 15-level z-stacks per time point. Hybrid architecture combining Siamese Networks, ConvLSTM, and Attention U-Net achieves **Dice score: 84.62%**, outperforming U-Net and SAM baselines in low-contrast conditions. Quantitative invasion analysis reveals radial leader–follower dynamics and morphological dissociation from classical EMT assumptions. *Manuscript in preparation.*
- **Interpretable image-based phenotyping.** ResNet-18 + Grad-CAM pipeline for non-destructive quality assessment of 3D cardiac spheroids from bright-field microscopy (86.5% accuracy); saliency maps highlight boundary and internal morphological patterns consistent with expert annotation, providing mechanistic interpretability beyond classification. *Manuscript under review, Biofabrication (submitted 2025).*
- **Weakly supervised differentiation QC.** Deep learning framework for automated NK-cell maturation assessment from Day-10 bright-field EB images; domain-specific preprocessing (Otsu-based ROI, circular soft masking), imbalance-aware training (class-weighted loss, Mixup, label smoothing), and stratified group 5-fold cross-validation. ResNet-18 achieves **96.25% accuracy (AUC: 0.99)**, with t-SNE confirming discriminative morphological representations. First-author paper, *IEEE CBMS 2026 (accepted).*
- **Research infrastructure.** Modular PyTorch pipelines with deterministic seeding, configuration management, and experiment tracking; shared as reusable research infrastructure across Tu Lab projects.

Visiting Research Intern — TEEP Fellowship — May 2024 – Oct 2024
National Cheng Kung University, Tu Lab · Advisor: Prof. Ting-Yuan Tu — *Tainan, Taiwan*

- **Longitudinal spatiotemporal organoid modeling.** Designed a temporally-aware longitudinal imaging pipeline to predict downstream functional differentiation outcomes from early time-point images; introduced temporal augmentation strategies and multi-timepoint feature integration for early-stage phenotype forecasting. Methodological foundation of the IEEE CBMS 2026 framework.
- **Oral presentation.** "Early Prediction of iPSC Differentiation Quality via Deep Learning." TSBME & TSB Joint Conference, 2024 (invited).

Research Intern — IIPP Fellowship — Jan 2024 – Mar 2024
National Yang Ming Chiao Tung University · Advisor: Prof. Ray-Hua Horng — *Hsinchu, Taiwan*

- **Surrogate ML modeling for scientific instruments.** Feature-engineered regression pipeline for semiconductor growth-parameter prediction (FWHM; $R^2 = 0.80$) via cross-validated model selection; demonstrates scientific ML applied to physical measurement systems. Selected from ∼600 nationwide applicants (NTD 30,000/month stipend).

Publications & Presentations

[1] K. Akanksha et al. "Deep Learning for Automated Quality Assessment of NK Cell Differentiation in iPSC Cultures." *IEEE CBMS 2026. (First author, accepted.)*

[2] K. Akanksha et al. "Early Prediction of Differentiation Quality in Cardiac Spheroids Using Interpretable Deep Learning." *Biofabrication (under review, submitted 2025).*

[3] K. Akanksha. "Early Prediction of iPSC Differentiation Quality via Deep Learning." *Invited oral presentation, TSBME & TSB Joint Conference, 2024.*

Selected Research Projects

Siamese Re-ID for Live-Cell Tracking in 3D Cancer Invasion

Longitudinal Temporal Model for Organoid Maturation Prediction

PyTorch · contrastive learning · ConvLSTM · Attention U-Net · confocal time-lapse
temporal augmentation · multi-timepoint features · early prediction · bright-field microscopy

Technical Skills

Programming & tools: Python (primary), SQL, shell scripting, Linux research workflows; Git/GitHub, Docker.

Deep learning & scientific ML: PyTorch, TensorFlow, Keras, scikit-learn; CNNs (ResNet, DenseNet, MobileNet), Siamese/metric-learning networks, ConvLSTMs, Attention U-Net, Vision Transformers (working knowledge), transfer learning, weakly supervised training; self-supervised methods (active self-study).

Computational imaging & biological CV: time-lapse microscopy analysis (bright-field, confocal), z-stack reconstruction, CLAHE contrast enhancement, ROI detection (Otsu thresholding), circular soft masking, multi-object tracking, instance segmentation, image-based phenotyping, Grad-CAM/saliency interpretability, t-SNE feature visualization.

Experimental & statistical rigor: stratified group cross-validation, class-imbalance handling, Mixup augmentation, label smoothing, deterministic seeding, batch-aware experiment tracking.

Data & visualisation: Pandas, NumPy, Matplotlib, Plotly; statistical inference; Tableau, Power BI.

Awards & Honors

- **Top of Master's Cohort**, NCKU Biomedical Engineering (2025–Present) — highest GPA in cohort (4.24/4.30).
- **TEEP Research Fellowship**, Ministry of Education, Taiwan (2024) — competitive fellowship for deep-learning research at NCKU prior to graduate enrollment.
- **IIPP International Research Fellowship**, NYCU (2024) — selected from ∼600 applicants nationwide; NTD 30,000/month.
- **Merit-cum-Means Scholarship**, Delhi Government — awarded for academic excellence during undergraduate studies.
- **Invited Oral Talk**, TSBME & TSB Joint Conference (2024) — early prediction of iPSC differentiation quality via interpretable deep learning.

Relevant Coursework

Computer Vision and Deep Learning (A+, NCKU)	Introduction to Digital Health and AI in Medical Applications (A+, NCKU)	Adv
Statistics and Clinical Trials for Medical Devices (A+, NCKU)	Deep Learning (DSEU)	Adv
Time Series and Forecasting (DSEU)	Mathematical Foundation for Data Analytics (DSEU)	Com

Service & Outreach

- **President, Indian National Students Association (INSA), NCKU** (Jul 2025–Present) — lead 100+ member community; coordinate academic and intercultural programming with the NCKU International Affairs Office.
- **Mentor, Desh ke Mentor, Delhi Government** (Oct 2022–Apr 2023) — mentored high-school students on academics and career planning.

Languages & Interests

Languages: English (fluent) · Hindi (native) · Mandarin Chinese (basic) · Sanskrit (basic).

Interests: Badminton · Reading · Travel & hiking · Photography · Music.